Surface Multi-Layer

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March 23, 2026

Keypoints

- Enhancing reduced-gravity model reconstruction in the context of data assimilation by acknowledging the truncation error
- Introducing a physics-derived regularization term for the cost function to enforce multilayer QG dynamics
- Improving the robustness of the model by introducing a deformation radius correcting parameter in the control vector

1 Introduction

In physical oceanography, the majority of available data is acquired by satellites that cannot measure the interior ocean state. The recent launch of the Surface Water and Ocean Topography (SWOT) mission (Fu et al., 2024) has paved the way for sea surface reconstruction with unprecedented resolution. However, due to the shape of its orbit, this satellite covers the entire surface of the globe in 21 days. Although most parts of the globe are visited multiple times within this 21d-period, many gaps remain in the collected data. The reconstruction of sea-surface dynamics through partial observation has been investigated for a long time. In the late 1980s, some physics-motivated flow reconstruction methods were developed, either focusing on surface (Cong & Ikeda, 1995; Cong et al., 1998; Cooper & Haines, 1996; Holland & Malanotte-Rizzoli, 1989; Mey & Robinson, 1987) or subsurface (Haines, 1991; Hurlburt, 1986) flow. In the 1990s, statistical methods such as optimal interpolation (Rienecker & Miller, 1991; Traon et al., 1998) emerged, forming the basis of the DUACS product which remains a widely used real-time gridded ocean product. This interpolation method uses only observation data to reconstruct surface ocean state.

To make use of theoretical models of ocean surface dynamics, data assimilation (DA) techniques started to be applied to the ocean system. In order to reduce reconstruction errors, DA combines both simulated and observed data while accounting for their specific sources of error (Lahoz et al., 2010). A wide variety of methods have been developed and applied to the reconstruction of surface ocean dynamics (Martin et al., 2025). In DA methods, so-called reduced-order models are used to propagate in time the system state. These models use simplified frameworks to emulate ocean system dynamics, allowing to perform efficiently complex DA algorithms. Around the midlatitudes, a simplified framework of choice if the quasi-geostophic (QG) theory. This framework, developed in the 1950 (Charney & Phillips, 1953; Charney et al., 1950), describes the behavior

of flows around the geostrophic balance through the material conservation of potential vorticity (PV) by the geostrophic flow. Such theory has paved the way for many simplified numerical models, whether atmospheric or oceanic (Held et al., 1995; Phillips, 1951). Although it describes 3D dynamics, it can be discretized by stacking layers of fluid with constant vertical properties, yielding the multilayer QG framework (Vallis, 2006). The ability of such framework in reconstructing sublayer flow has been widely investigated in the late 1980s (Haines, 1991; Hurlburt, 1986), but satisfying reconstructions are achieved after numerous surface observations have assimilated into the models. Regarding surface flow reconstruction-only, assimilation into a two-layer QG model has been experimented by Cong et al., 1998 and showed encouraging results. However, the lack of subsurface direct observations (and thus of precise estimation for initial sublayer flow state) has motivated the use of reduced-gravity (RG) QG model in data assimilation. This framework is directly derived from the multilayer one by considering deep ocean flow to be at rest. This additional simplification offers an evolution equation for the surface flow without requiring any information on subsurface state. These specific characteristics naturally led to the choice of the RG framework for modeling surface dynamics.

Using a RG model, Ubelmann et al., 2015 presented the so-called dynamical interpolation method that consist in refining the optimal interpolation reconstruction by using a RG model to benefit from it ability to render non-linear dynamics. This method has been able to improve the reconstruction of the flow (Ubelmann et al., 2016). In order to account for any source of error in the RG model, the VarDyn framework (Le Guillou et al., 2023, 2025) introduces an external forcing into the RG advection equation. Such forcing is then estimated through variational assimilation to approach the rich and complex dynamics of the ocean surface. However, in both methods remains the approximation that the flow is at rest below the ocean surface.

Motivated by a rewriting of the top layer QG equations, we tackle truncation error through a parameterization of the sublayer stream function. We introduce an external forcing in the RG equations which represents the advection of sublayer stream function by surface flow. In a context of partial surface observations, variational assimilation is then performed to find the sublayer field that allows the most accurate reconstruction of the surface flow. As our method provides a direct estimation for sublayer stream function, we are also able to replace the usual background term of the loss function by a physics-based regularization enforcing multilayer QG dynamics of the surface flow. Method validation is then performed through two experiments: comparison to a three-layer QG model to monitor truncation-only error mitigation, and comparison to a primitive equation simulation to assess the model's reconstruction capabilities of a richer dynamics flow.

2 The multilayer QG theory

QG dynamics relies on the material conservation of potential vorticity (PV) by geostrophic flow. As PV is related to the stream function through an elliptic operator, the full geostrophic flow can be recovered from the PV field through the inversion of such operator (Hoskins et al., 1985). Multilayer QG, the vertical discretization of the continuously stratified QG equation, is obtained by stacking layers of fluid with constant vertical properties (Pedlosky, 1987; Vallis, 2006). In the following we will recall these equations in a N-layer fluid. Using $\underline{\cdot}$ to represent layered fields ($\underline{\chi} = [\chi_1, \dots, \chi_N]^\top$), we denote

- $\underline{\rho}$ the reference layer densities,
- $\underline{H}$ the reference layer thickness,
- $\underline{g}$ the reduced gravities ($g_i = g\left(\rho_i - \rho_{i-1}\right)/\rho_1$, with the convention $g_1 = g$),
- $\underline{\psi}$ the stream function vector
- $\underline{q}$ the PV vector

Introducing the horizontal gradient $\boldsymbol{\nabla} = \begin{bmatrix}\frac{\partial}{\partial x} & \frac{\partial}{\partial y}\end{bmatrix}^\top$ along with the perpendicular horizontal gradient $\boldsymbol{\nabla}^\perp = \begin{bmatrix}-\frac{\partial}{\partial y} & \frac{\partial}{\partial x}\end{bmatrix}^\top$, the free-surface multilayer QG equations (in the absence of external forcing) in a layer i read

$$q_i = \Delta\,\psi_i - f_0^2(\mathsf{A}\underline{\psi})_i, \tag{1a}$$

$$\frac{\partial q_i}{\partial t} + \boldsymbol{\nabla}^\perp\,\psi_i \cdot \boldsymbol{\nabla}\,q_i = 0, \tag{1b}$$

where the stretching matrix A, is given by

$$\mathsf{A} = \begin{bmatrix} \frac{1}{H_1 g_1} + \frac{1}{H_1 g_2} & -\frac{1}{H_1 g_2} & \cdot & \cdot \\ -\frac{1}{H_2 g_2} & \frac{1}{H_2 g_2} + \frac{1}{H_2 g_3} & -\frac{1}{H_2 g_3} & \cdot \\ \vdots & \ddots & \ddots & \vdots \\ \cdot & \cdot & -\frac{1}{H_N g_N} & \frac{1}{H_N g_N} \end{bmatrix}. \tag{2}$$

Direct observations through satellite sensing are only able to measure sea surface height (SSH), which can be matched to top layer stream function ψ_1. Remote sensing alone can not provide direct measurement on the sublayer flow ψ_2. However, the top layer PV (q_1), the dynamics of which govern surface flow, depends on the stream function from layers 1 and 2 as

$$q_1 = \Delta\,\psi_1 - \frac{f_0^2}{H_1 g_1}\psi_1 - \frac{f_0^2}{H_1 g_2}\left(\psi_1 - \psi_2\right). \tag{3}$$

To overcome such limitation, the RG framework considers the sublayer to be at rest, hence $\psi_2 = 0$. The corresponding top-layer PV q^{RG} expression yields

$$q^{RG} = \Delta\,\psi_1 - f_0^2\left(\frac{1}{H_1 g_1} + \frac{1}{H_1 g_2}\right)\psi_1, \tag{4}$$

which is advected by ψ_1, hence,

$$\frac{\partial q^{RG}}{\partial t} + \boldsymbol{\nabla}^\perp\,\psi_1 \cdot \boldsymbol{\nabla}\,q^{RG} = 0. \tag{5}$$

This is the formulation of the RG model which was used in the first time to perform dynamical interpolation (Ubelmann et al., 2015, 2016). Benefiting from the ability of the

QG model to generate non-linear dynamics, such method has been able to outperform statistics-based interpolation. However, due to the truncated QG equations, significant errors can emerge when performing optimal interpolation using solely the RG model, as pointed out in Manucharyan et al., 2021. In order to address this limitation, the VarDyn framework (Le Guillou et al., 2023, 2025) acknowledges the model error by introducing an external forcing, ϵ, into the advection equation,

$$\frac{\partial q^{RG}}{\partial t} + J\left(\psi_1, q^{RG}\right) = \epsilon. \tag{6}$$

However, given the QG PV definition, the top layer PV can be written

$$q_1 = q^{RG} + \frac{f_0^2}{H_1 g_2}\psi_2. \tag{7}$$

Using this decomposition into the top layer PV advection equation yields

$$\frac{\partial q^{RG}}{\partial t} + J\left(\psi_1, q^{RG}\right) = -\frac{f_0^2}{H_1 g_2}\left[\frac{\partial \psi_2}{\partial t} + \boldsymbol{\nabla}^{\perp}\psi_1 \cdot \boldsymbol{\nabla}\psi_2\right] \tag{8a}$$

$$= -\frac{f_0^2}{H_1 g_2}\frac{\mathrm{D}\psi_2}{\mathrm{D}t}. \tag{8b}$$

From this expression, we note that the usual RG model advection matches the multilayer formulation when $\mathrm{D}_t\psi_2 = 0$. Addressing the truncation error of the RG model resumes to emulating the right-hand side of eq. (8). Since, ψ_1 is the prognostic variable of the model, computing $\mathrm{D}_t\psi_2$ boils down to estimating ψ_2. Denoting $\tilde{\psi}_2$ such estimation, we define the corrected RG advection by

$$\frac{\partial q^{RG}}{\partial t} + J\left(\psi_1, q^{RG}\right) = -\frac{f_0^2}{H_1 g_2}\frac{\mathrm{D}\tilde{\psi}_2}{\mathrm{D}t}. \tag{9}$$

In the following, this model will be referred to as SurfML (Surface Multi-Layer). Additionaly to explicitly addressing the RG-related truncation error, this method also provides a proxy for the sublayer stream function, through $\tilde{\psi}_2$. Our specific parametrization of $\tilde{\psi}_2$ will be detailed in section 4. In the forthcoming, all numerical experiments were conducted using a Pytorch implementation of the multilayer QG equations derived from Thiry et al., 2024 (extended to withstand non-zero boundary conditions).

3 Variational assimilation

We consider a non-linear, time-evolving system whose state at time t is given by x_t^{true}. The observation operator $\mathcal{H}$ allows gathering a (discrete) observation $\mathbf{y}_t = \mathcal{H}\left(x_t^{\text{true}}\right)$ on the system state at time t. We introduce the propagator $\mathcal{M}$, which provides a (discrete) proxy $\mathbf{x}_t = \mathcal{M}\left(\theta \mid t\right)$ for x_t^{true}, given the control vector θ which may include any parameter the propagator dynamics depends on (initial and boundary condition, model parameters...). Observations drawn from the true state can be compared to their counterpart generated

using $\mathcal{H} \circ \mathcal{M}$. Given a set of observations $(\mathbf{y}_i)_{i \in \{1,\dots,N_{\text{obs}}\}}$ evaluated at times $(t_i)_{i \in \{1,\dots,N_{\text{obs}}\}}$, the performance of $\mathcal{M}$ in reproducing the evolution of the system can be evaluated by the observation loss

$$\mathcal{J}_{\text{obs}}(\theta) = \sum_{i=1}^{N_{\text{obs}}} \|\mathbf{y}_i - \mathcal{H} \circ \mathcal{M}(\theta \mid t_i)\|^2_{\mathsf{R}^{-1}}, \tag{10}$$

with the observation error covariance matrix R. Variational assimilation reduces to finding

$$\theta^\star = \arg\min_\theta \mathcal{J}_{\text{obs}}(\theta). \tag{11}$$

However, there are no guarantees that enough data is available for the minimization problem to be well constrained. Therefore, provided a prior estimate of the control vector, θ^b, and its error covariance matrix, B, the cost function is usually regularized by penalizing the deviation of θ from θ^b. Introducing the regularization loss

$$\mathcal{J}_{\text{reg}}(\theta) = \|\theta - \theta^b\|^2_{\mathsf{B}^{-1}}, \tag{12}$$

the cost function rewrites

$$\mathcal{J}(\theta) = \mathcal{J}_{\text{obs}}(\theta) + \mathcal{J}_{\text{reg}}(\theta). \tag{13}$$

Supposing we can estimate the gradient $\boldsymbol{\nabla} \mathcal{J}$ of the loss function, minimization can be performed through gradient descent. Regarding our specific experimentations, since all models and operation are implemented using Pytorch, we benefit from the library's built-in automatic differentiation (Paszke et al., 2017) to both retrieve loss function gradient and perform a descent algorithm.

4 The forward model

Hereafter we detail the specificities of the SurfML model and of the sublayer parameterization which was used as propagator in the variational assimilation algorithm.

4.1 Sublayer parameterization

In order to keep a reasonable number of parameters for the parameterization of $\tilde{\psi}_2$, we use a reduced-order space to represent its value. We have decided to express $\tilde{\psi}_2$ over a quite simple space so that SurfML performances could not be attributed to the structure of the parameterization rather than the actual error model. Therefore, the spatio-temporal behavior of $\tilde{\psi}_2$ is prescribed through a partition of unity of Gaussian envelopes, both in time and space. Denoting N_T the number of time-related envelopes and N_{XY} the number of space-related envelopes at a given time i, for given $i \in \{1, \dots, N_T\}$ and $j \in \{1, \dots, N_{XY}\}$, we define

$$E_i^T(t) = \exp\left[-\left(\frac{t-t_i}{\sigma_T}\right)^2\right], \tag{14a}$$

$$E_j^{XY}(x,y) = \exp\left[-\left(\frac{x-x_j}{\sigma_X}\right)^2\right]\exp\left[-\left(\frac{y-y_j}{\sigma_Y}\right)^2\right], \tag{14b}$$

where t_i, x_j, y_j are the locations of the envelope centers and $\sigma_T, \sigma_X, \sigma_Y$ their standard deviation. We also define $E^T = \sum_{i=1}^{N_T} E_i^T$ and $E^{XY} = \sum_{i=1}^{N_{XY}} E_j^{XY}$. Using these notations, the overall decomposition of $\tilde{\psi}_2$ at time t and location (x,y) reads

$$\tilde{\psi}_2(t,x,y) = \frac{\sum_{i=1}^{N_T}\sum_{j=1}^{N_{XY}} c_{i,j} E_i^T(t)\, E_j^{XY}(x,y)}{E^T(t)\, E^{XY}(x,y)}, \tag{15}$$

where the $\left(c_{i,j}\right)_{i\in\{1,\dots,N_T\}, j\in\{1,\dots,N_{XY}\}}$ are coefficients inferred through cost function minimization. Setting all coefficient to 0 restores the dynamics of a flow following the RG assumption and is chosen as the initial condition for the parameterization. Envelopes are regularly spaced over both time and space and their standard deviations are chosen so that consecutive envelopes cross at $-3\,\mathrm{dB}$. Hence, considering consecutive time-related (resp., space-related) envelopes to be distant from Δt (resp., Δx and Δy), their standard deviations are given by

$$\sigma_T = \frac{\Delta t}{\sqrt{2\ln 2}}, \tag{16a}$$

$$\sigma_X = \frac{\Delta x}{\sqrt{2\ln 2}}, \tag{16b}$$

$$\sigma_Y = \frac{\Delta y}{\sqrt{2\ln 2}}. \tag{16c}$$

Such parameterization allows representing a smooth field while enabling the emergence of some rather small scale patterns.

4.2 Physics-based regularization

Following eq. (1b), PV advection in the layer 2 reads

$$\partial_t q_2 + \boldsymbol{\nabla}^\perp \psi_2 \cdot \boldsymbol{\nabla} q_2 = 0. \tag{17}$$

However, as the shape of our model error yields a proxy for ψ_2, we define the sublayer PV using the truncation of its multilayer definition (we consider $\psi_3 = 0$), retrieving the following proxy for q_2,

$$\tilde{q}_2 = \Delta\,\tilde{\psi}_2 - \frac{f_0^2}{H_2 g_2}\left(\tilde{\psi}_2 - \psi_1\right). \tag{18}$$

We can compute an estimation of the sublayer advection residual with $\partial_t \tilde{q}_2 + \boldsymbol{\nabla}^\perp \tilde{\psi}_2 \cdot \boldsymbol{\nabla} \tilde{q}_2$. In order to enforce physical constraints during the optimization process, we introduce a physics-based regularization,

$$\mathcal{J}_{\text{reg}}\left(\psi_1, \tilde{\psi}_2\right) = \left\|\partial_t \tilde{q}_2 + \boldsymbol{\nabla}^\perp \tilde{\psi}_2 \cdot \boldsymbol{\nabla} \tilde{q}_2\right\|^2, \tag{19}$$

and we define the overall cost function with

$$\mathcal{J}\left(\tilde{\psi}_2\right) = \mathcal{J}_{\text{obs}}\left(\tilde{\psi}_2\right) + \gamma \mathcal{J}_{\text{reg}}\left(\tilde{\psi}_2\right), \tag{20}$$

where γ is an hyperparameter driving the relative importance of the regularization over the observation loss. Replacing the usual background state related regularization by this physics-based formulation is inspired by physics-informed neural networks (PINN) methodology (Raissi et al., 2019) and PINN-derived data assimilation approaches have proven the relevancy of regularizing using physical constraint (Oh, 2025; Ugur & Zhou, 2025).

Additionally, switching to a physics-based regularization removes the need for an a priori guess of the parameter. Since our method is based on reconstructing a proxy for the sublayer stream function, retrieving a non-trivial estimation for ψ_2 remains challenging. A possible estimation would have been to penalize $\tilde{\psi}_2$ from deviating from 0, thus preventing the model from stirring away from the RG framework. It would have also been possible to enforce a barotropic flow by penalizing deviation from ψ_1. However, both of these options pull the model toward one-layer dynamics whereas the physics-derived regularization allows for two-layer-like dynamics while still allowing the system to fall back to $\tilde{\psi}_2 \simeq 0$ or $\tilde{\psi}_2 \simeq \psi_1$ when appropriate.

It is also worth emphasizing that introducing this regularization does not strictly enforce the conservation of PV in the layer. It guides the model towards finding an expression for $\tilde{\psi}_2$ that approaches the conservation $\tilde{q}_2$. However, as there are no guarantees that $\tilde{q}_2$ should be strictly conserved ($\tilde{\psi}_2$ is not exact and $\tilde{q}_2$ is computed through the truncation of the multilayer equations), this formulation allows for some flexibility on said conservation.

Furthermore, if we were to assimilate an initial condition ψ_2^0 for the bottom layer stream function in two-layer QG model, constraining the minimization problem would yield the question of finding a background estimation for ψ_2^0. Once again, two natural choices would be the following: considering 0 or ψ_1^0 (or a linear combination of both, as done by Cong and Ikeda, 1995), hence leaning toward either a surface intensified or a barotropic initial condition for the flow. As the background term would strongly call back the ψ_2^0 toward one of these options, the overall dynamics would be notably impacted by this decision. Indeed, according to Hurlburt, 1986, instantiating a two-layer model by setting the initial sublayer field at rest yields not better results regarding surface forecast than a regular RG model, thus emphasizing the need for a precise a priori estimation of sublayer initial state. Our regularization however, does not constrain the value of the sublayer stream function itself but rather its dynamics: it has to somehow satisfy a pseudo-advection of PV in the bottom layer, but as for its value it remains unconstrained. Additionally, this method allows for more flexibility regarding subsurface flow: if improving surface dynamics requires drifting away from the advection of the sublayer (truncated) PV (through the introduction of non-QG patterns in the reconstructed field for example), it can do so. A

strictly enforced two-layer QG model would not be able to do so without the introduction of an external forcing term in the sublayer advection equation.

4.3 Deformation radius correction

Other than the truncation error, incorrect the model stratification parameters (in the H_i or the g_i) could result in mismatch in the surface flow dynamics. Key characteristics of the flow dynamics are the so-called barotropic and first baroclinic deformation radii, which, in the multilayer QG framework, can be computed from the eigenvalues of the stretching matrix A. In order to increase the robustness of our two-layer-like model, we wish to allow for baroclinic deformation radius adjustments. Given the diagonalization of the two-layer stretching matrix,

$$\mathsf{A} = \mathsf{C}_{\mathrm{M2L}} \begin{bmatrix} \lambda_{\rightleftarrows} & 0 \\ 0 & \lambda_{\rightrightarrows} \end{bmatrix} \mathsf{C}_{\mathrm{L2M}}, \tag{21}$$

where $\lambda_{\rightleftarrows}$ is the baroclinic-related eigenvalue and $\lambda_{\rightrightarrows}$ the barotropic one, and $\mathsf{C}_{\mathrm{M2L}}$ (resp., $\mathsf{C}_{\mathrm{L2M}}$) the modes to layers (resp., layers to modes) passage matrix. The baroclinic radius correction is performed by introducing the parameter α, and the modified stretching matrix,

$$\tilde{\mathsf{A}}_\alpha = \mathsf{C}_{\mathrm{M2L}} \begin{bmatrix} (1+\alpha)\,\lambda_{\rightleftarrows} & 0 \\ 0 & \lambda_{\rightrightarrows} \end{bmatrix} \mathsf{C}_{\mathrm{L2M}}, \tag{22a}$$

$$= \mathsf{A} + \alpha\lambda_{\rightleftarrows} \mathsf{C}_{\mathrm{M2L}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathsf{C}_{\mathrm{L2M}}, \tag{22b}$$

which coincides with the default stretching matrix when $\alpha = 0$. The barotropic deformation radius then remains unchanged as it is linked to the total depth which can be quite easily approximated whereas the baroclinic deformation radius, $L_{\rightleftarrows}$ becomes

$$L_{\rightleftarrows} = \frac{1}{f_0\sqrt{(1+\alpha)\,\lambda_{\rightleftarrows}}}. \tag{23}$$

Our model implementation replaces A by $\tilde{\mathsf{A}}_\alpha$ in the usual two-layer QG equations and the parameter α is added in the control vector, granting one additional degree of freedom for the variational assimilation.

5 Method validation

We have considered two validation experiments. In the first one, we have set a double-gyre experiment using a three-layer QG model. Therefore, the reconstruction error from the RG model (dynamical interpolation situation) is mostly explained by the truncation of the QG equations. Within this framework, we can also assess the ability of SurfML model in reconstructing top layer dynamics and compare its performances with VarDyn. Two variants of this setup have been explored. One where model parameters (reference layer

depth, reduced gravity values…) are the same in the reference QG model and in the RG counterpart. This configuration also allows a fair comparison for $\tilde{\psi}_2$ and ψ_2. In the other variant, model parameters are modified to assess the robustness of each of the models when dealing with incorrect baroclinic deformation radius. The second validation experiment uses surface fields from the eNATL60 simulation (Brodeau et al., 2020) as the reference flow. This experiment investigate the capability of our model in reproducing a flow that present ageostrophic, complex and rich dynamical features.

In both experiments, observations are drawn from the surface using an observing system simulation experiment that simulates SWOT-like sampling tracks.

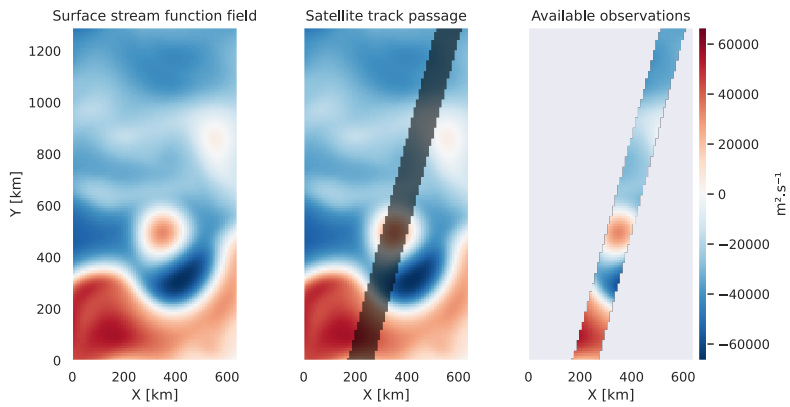

Figure 1: Example of simulated satellite track over the surface (Surface fields extracted from a three-layer QG double-gyre simulation).

As this work focuses on introducing a new parameterization to reduce model error, we have decided to only consider such parameterization in the control vector. Indeed, since all tested frameworks (RG, VarDyn, SurfML) share identical initial and boundary conditions, differences in performances only results from model error and its correction. Ideal initial and boundary conditions are therefore retrieved from the surface flow of the reference simulation. Furthermore, the estimation of initial and boundary conditions through variational assimilation has already been widely investigated (Ayoub, 2006; Cong et al., 1998; Gunson & Malanotte-Rizzoli, 1996; Seiler, 1993; Zou et al., 1995), suggesting that our methodology could quite easily be adapted to a more realistic data assimilation configuration.

Assimilation is conducted on 20 days-long periods. Over these periods, satellite passages are simulated and, when a track spans over the considered domain, observation loss is evaluated using the collected data. Regarding the SurfML model, regularization loss is evaluated at every time step. The VarDyn regularization however, is implemented so that wavelet scales are penalized following a -5/3 power law. Small scales are therefore penalized strongly than large scales.

Parameters for all models are detailed in the following table

Parameter	Value			
$\mathrm{d}t$	$7200\,\mathrm{s}$			
$\mathrm{d}x$	$10\,\mathrm{km}$			
$\mathrm{d}y$	$10\,\mathrm{km}$			
f_0	$9.375 \times 10^{-5}\,\mathrm{s}^{-1}$			
β	$1.754 \times 10^{-11}\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$			
Parameter	QG3L	RG	VarDyn	SurfML
H_1	$400\,\mathrm{m}$	$400\,\mathrm{m}$	$293\,\mathrm{m}$	$400\,\mathrm{m}$
H_2	$1100\,\mathrm{m}$	.	.	$1100\,\mathrm{m}$
H_3	$2600\,\mathrm{m}$	.	.	.
g_1	$9.81\,\mathrm{m/s^2}$	$0.0249\,\mathrm{m/s^2}$	$0.025\,\mathrm{m/s^2}$	$9.81\,\mathrm{m/s^2}$
g_2	$0.025\,\mathrm{m/s^2}$	.	.	$0.025\,\mathrm{m/s^2}$
g_3	$0.0125\,\mathrm{m/s^2}$	.	.	.

Table 1: Model parameters

VarDyn parameters have been chosen so that the one layer RG model matches the baroclinic deformation radius of a two-layer QG model. The value of this deformation radius is about 29 km, similar to the one described in Le Guillou et al., 2023, namely 30 km. The parameter of the RG model however are set so that the one-layer dynamics matches the one of the truncated QG model.

Model performances will be evaluated using the normalized Root Mean Squared Error (nRMSE). Given a field χ and its reference value χ_{ref}, the corresponding nRMSE is

$$\mathrm{nRMSE}\left(\chi, \chi_{\mathrm{ref}}\right) = \sqrt{\frac{\left\|\chi - \chi_{\mathrm{ref}}\right\|^2}{\left\|\chi_{\mathrm{ref}}\right\|^2}}. \tag{24}$$

The error normalization is such that evaluating the nRMSE of 0 over a reference field yields 1. A nRMSE greater then 1 indicates that predicting 0 would actually improve the score.

5.1 Multilayer QG

Since the reference simulation is forced by wind to display a double-gyre, we also include the wind as an external forcing within the advection equations for all models, denoting $\tau = \begin{bmatrix}\tau_x & \tau_y\end{bmatrix}^\top$ the wind stress tensor, the advection equation rewrites

$$\frac{\partial q^{RG}}{\partial t} + \boldsymbol{\nabla}^\perp \psi_1 \cdot \boldsymbol{\nabla}\, q^{RG} = F + \boldsymbol{\nabla}^\perp \cdot \tau \tag{25}$$

where F represent the additional forcing that depends on the frameworks,

$$F_{\mathrm{RG}} = 0, \tag{26a}$$

$$F_{\mathrm{VarDyn}} = \epsilon, \tag{26b}$$

$$F_{\mathrm{SurfML}} = \frac{-f_0^2}{H_1 g_2} \frac{\mathrm{D}\tilde{\psi}_2}{\mathrm{D}t}. \tag{26c}$$

As the reference was established over a 2560 km by 5120 km domain, we have decided to focus on subdomain of the said field. We selected four 640 km x 1280 km zones, Z1, Z2, Z3 and Z4

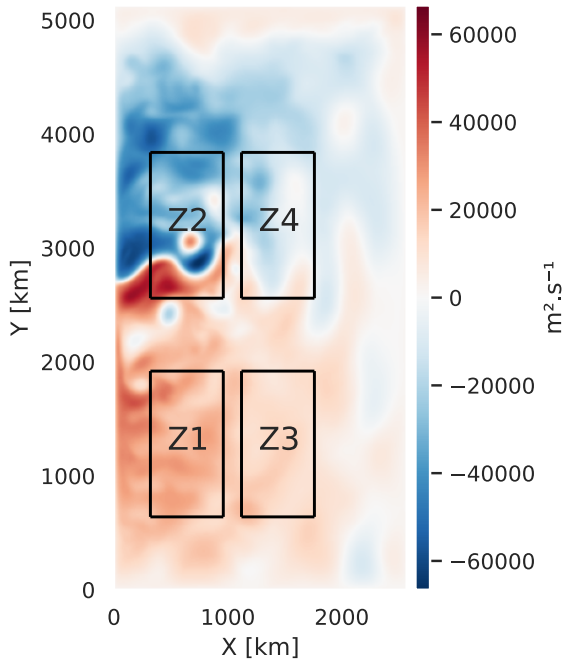

Figure 2: Detail of the 4 zones of the three-layer QG experiment.

In each of the four zones, the OSSE follows the same tracks, observing 13243 gridpoints. The VarDyn error formulation requires 8512 control parameters whereas the SurfML method uses 6322 control parameters (6321 for the reduced-order decomposition and 1 for the deformation radius correction). Both VarDyn and SurfML regularizations are scaling through an hyperparameter (γ). Its value has been fitted over the first period in Z2 and then reused in all areas and periods.

We will only display the results on Z2 and Z3 as they present different dynamical features. Z2 for instance is at the heart of the eastward jet and present a surface intensified dynamics. On Z3 however the baroclinic components of the flow is stronger. We thus consider that both zones can illustrate a satisfying variability of flow. Results (averaged on 3 consecutive 20-days periods) are shown in fig. 3.

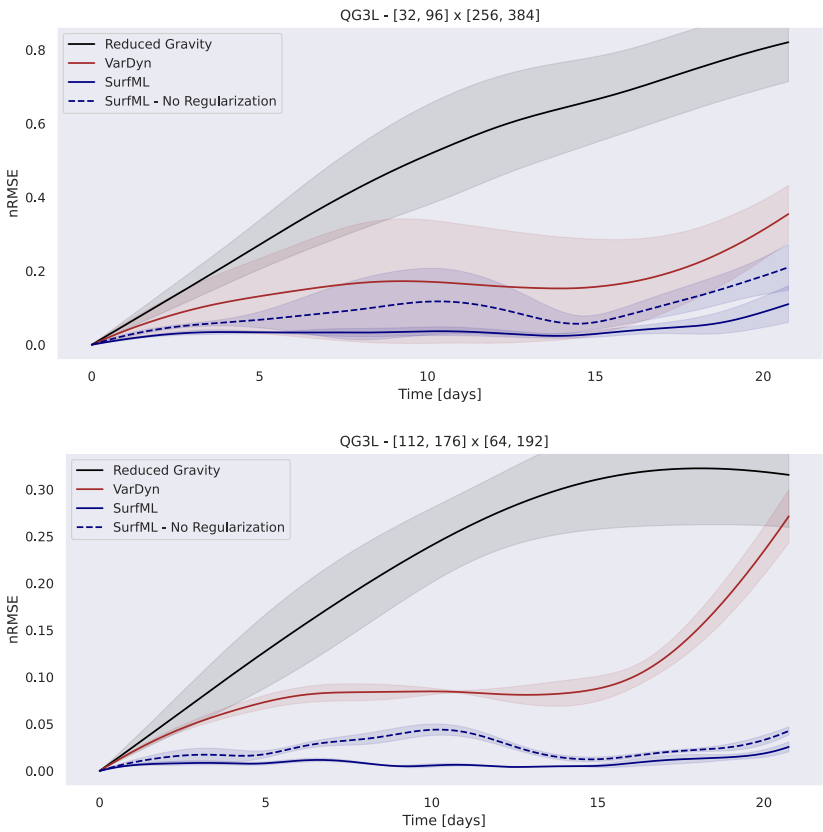

Figure 3: nRMSE on three-layer QG surface stream function (averaged over 3 consecutive 20-days periods) on Z2 (top) and Z3 (bottom). Lines represent the nRMSE average and the shaded area the standard deviation. Models are identified by the line color, black for RG, brown for VarDyn and blue for SurfML. The dashed line represents the non-regularized SurfML (hence $\gamma = 0$ in eq. (20)).

Results show that SurfML outperforms VarDyn in all considered areas and periods. Comparison between non-regularized and regularized SurfML highlights the benefits of the physics-based regularization. It is also worth mentioning that, as a comparison, we have tried fitting $\tilde{\psi}_2$ using direct (full-field) observation from ψ_2. Plugging the resulting parameters into the SurfML model yields similar (slightly worse) results as SurfML.

Add performance on U,V fields ?

5.1.1 Estimation of the sublayer stream function

As the SurfML method provides an estimation for the sublayer flow, ψ_2, we have evaluated the reconstruction of ψ_2 by $\tilde{\psi}_2$.

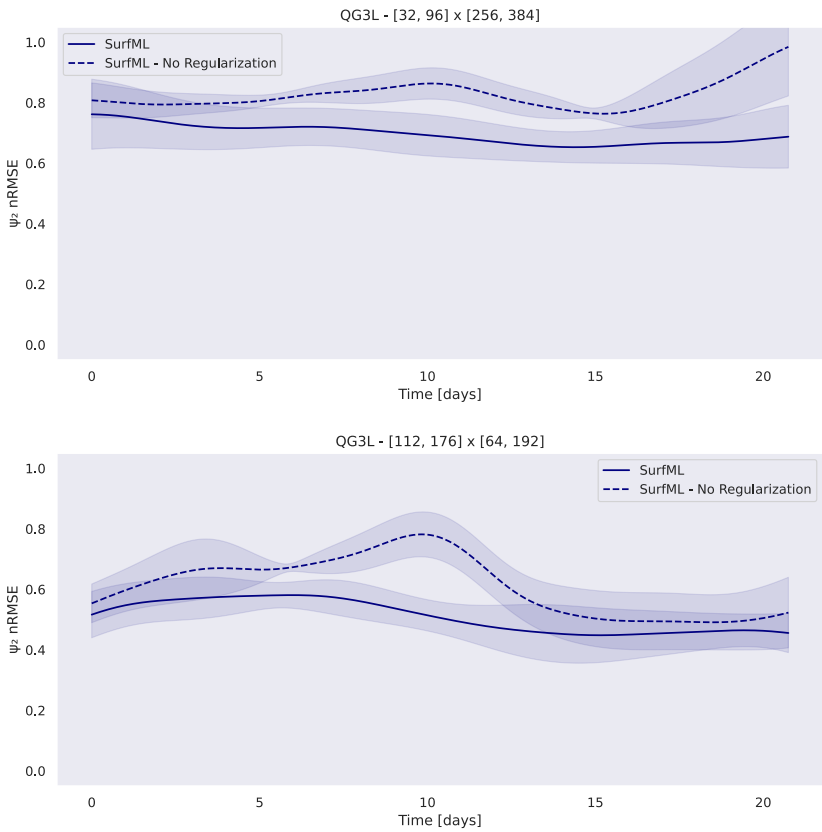

Figure 4: nRMSE on three-layer QG sublayer stream function (averaged over 3 consecutive 20-days periods) on Z2 (top) and Z3 (bottom). SurfML model is identified by the solid blue line and non-regularized SurfML by the dashed blue line.

Although results show that the regularization improves the reconstruction of ψ_2 by $\tilde{\psi}_2$, the quality of the overall reconstruction is far from being as it was for the surface flow. Earlier attempts, focusing on subsurface flow reconstruction, were able to reach lower nRMSE values but only after more than 60 days of model spin-up under regular surface state assimilation (Haines, 1991; Hurlburt, 1986). One additional detail regarding the SurfML is that the model formulation actually never uses the actual $\tilde{\psi}_2$ field in the advection but rather its derivatives, through $\mathrm{D}_t\tilde{\psi}_2$, which are therefore better-constrained than $\tilde{\psi}_2$. Hence, we look at the reconstruction of $\partial_t\psi_2$ by $\mathrm{D}_t\tilde{\psi}_2$.

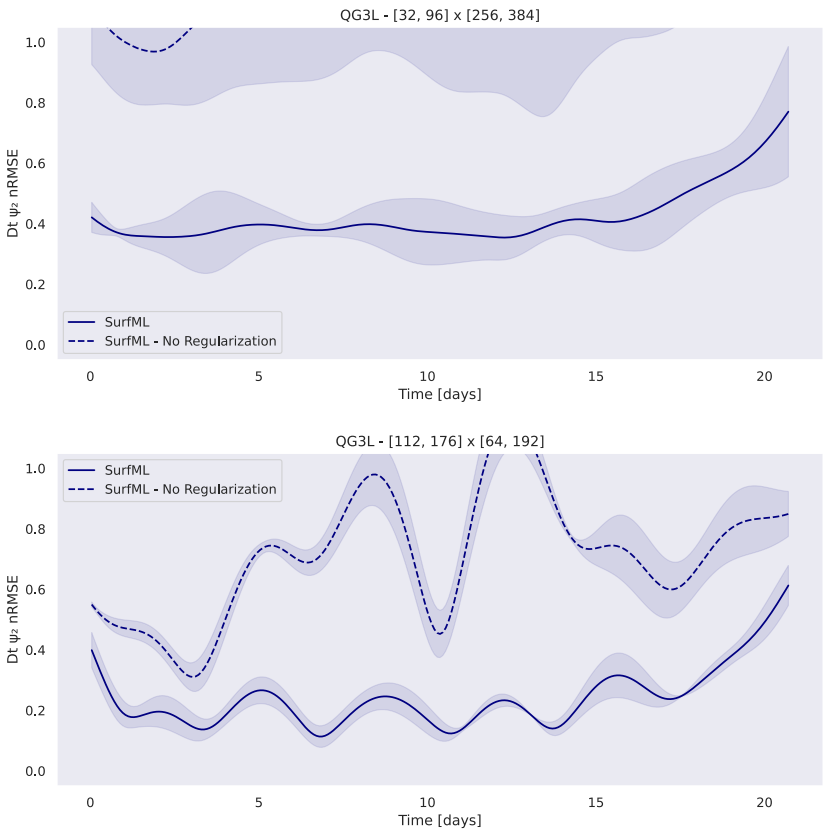

Figure 5: nRMSE on material derivative of three-layer QG sublayer stream function (averaged over 3 consecutive 20-days periods) on Z2 (top) and Z3 (bottom). SurfML model is identified by the solid blue line and non-regularized SurfML by the dashed blue line.

Although still not as good as top layer reconstruction scores, sublayer stream function material time derivative is better reconstructed than the field itself. Actually, when projecting the sublayer stream function onto the reduced-order space, the resulting error in the material derivative of the sublayer stream function is not very far from the one obtained using SurfML. Once again, result show the benefit of using the physics-based regularization.

5.1.2 Robustness

To assess the relevancy of the deformation radius correction presented in section 4.3, we keep the same parameters for the three-layer QG reference model, and we modify the other models. Modified parameters are detailed in the following table

Parameter	QG3L	RG	VarDyn	SurfML
H_1	400 m	400 m	360 m	600 m
H_2	1100 m	.	.	900 m
H_3	2600 m	.	.	.
g_1	9.81 m/s^2	0.002 499 m/s^2	0.0025 m/s^2	9.81 m/s^2
g_2	0.025 m/s^2	.	.	0.0025 m/s^2
g_3	0.0125 m/s^2	.	.	.

Table 2: Model parameters in the pertubed framework

We then conduct the same experiment as before and monitor the ability of the different models to tamper incorrect parameters and reconstruct the surface dynamics.

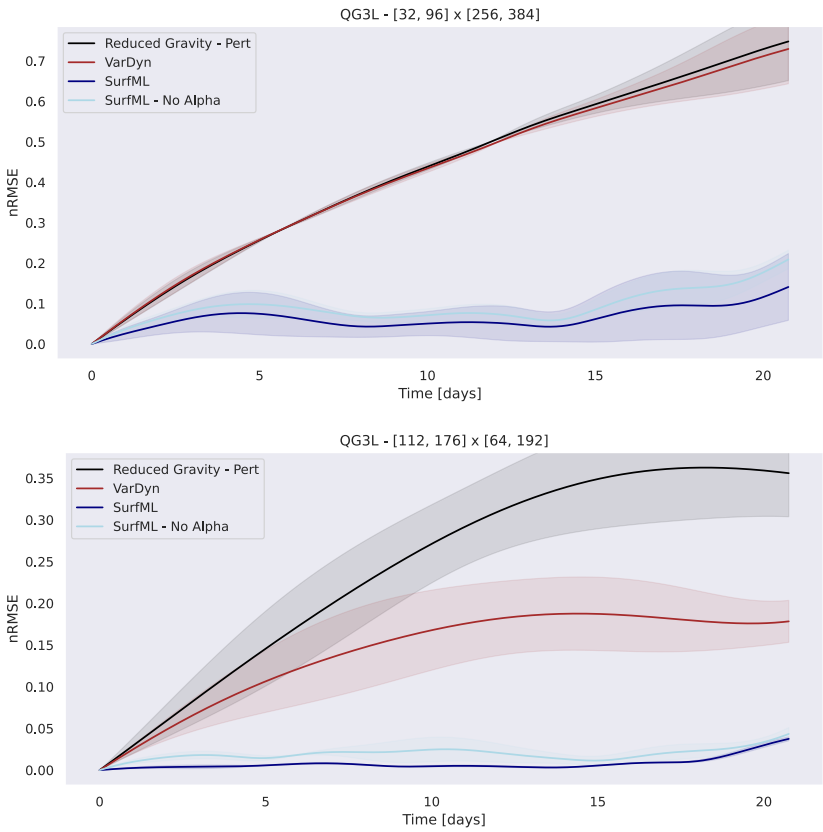

Figure 6: nRMSE on three-layer QG surface stream function (average over 3 consecutive 20-days periods) on Z2 (top) and Z3 (bottom) using perturbed models.Lines represent the nRMSE average and the shaded area the standard deviation. Models are identified by the line color, black for RG, brown for VarDyn and blue for SurfML. A light blue line is used for the SurfML model with $\alpha = 0$ (hence no deformation radius correction).

VarDyn performances greatly degrades when perturbing stratification parameters. This can be explained by the fact that all basis parameter has been carefully tuned using a 30 km deformation radius for the model.. SurfML however easily handles perturbation of the stratification parameters. The introduction of a single additional parameter in the control vector, α, significantly reduces the nRMSE.

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5.2 eNATL60

Using field from the eNATL60 simulation (Brodeau et al., 2020), we monitor SurfML's reconstruction of a fluid with ageostrophic dynamics. We have focused on a single area of dimension 680 km x 1030 km centered on −55°N 40°E. Parameters for all models are detailed are the exact same as the ones from table 1, apart from Coriolis parameters which were set to $f_0 = 9.373\,\mathrm{s}^{-1}$ and $\beta = 1.753\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$. Surface stream function is inferred from eNATL60's SSH values. First, SSH anomaly $SSH - \langle SSH \rangle_{T,XY}$ is computed by removing the time-space average (evaluated over the 20 days period) from the SSH field. We then use the Drakkar Forcing Set 5.2 (Dussin et al., 2016) to retrieve the atmospheric pressure anomaly $p_{\mathrm{atm}} - \langle p_{\mathrm{atm}} \rangle_{XY}$, ($\langle \cdot \rangle_{XY}$ denoting the spatial-average operator). Finally, surface stream function Ψ verifies

$$\Psi_{\mathrm{surf}} = \frac{1}{f_0 \rho_0} \left(p_{atm} - \langle p_{\mathrm{atm}} \rangle_{XY} \right) + \frac{g}{f_0} \left(SSH - \langle SSH \rangle_{T,XY} \right), \tag{27}$$

where ρ_0 is the reference density for water.

As the resulting stream function contain small-scale features, it is then smoothed using a Gaussian filter and then downsampled from the high-resolution, curvilinear coordinates of the eNATL60 simulation into the regular, Cartesian grid of the QG models. It is this smoothed, down-scaled field that is used for both initial and boundary conditions of the QG models. The observations however are sampled from the non-filtered surface stream function field. Once again variational assimilation is conducted over 20-days periods and the quality of the reconstruction is assessed by using nRMSE to compare reference stream function to its reconstructed counterpart.

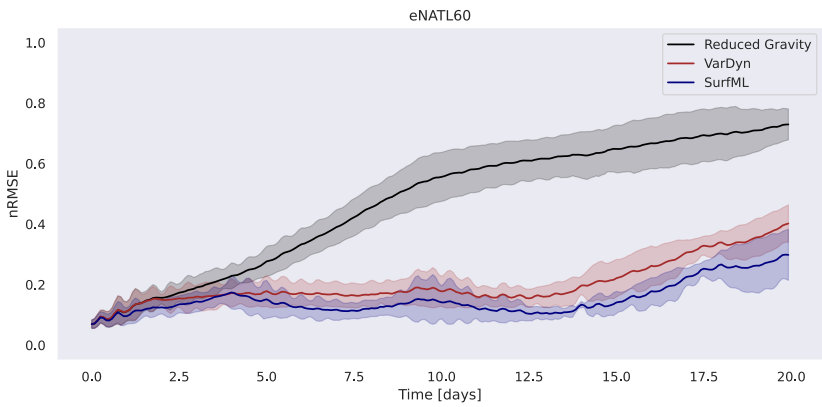

Figure 7: nRMSE on eNATL60 surface stream function (averaged over 3 consecutive 20-days periods). Models are identified by the line color, black for RG, brown for VarDyn and blue for SurfML.

Results show that SurfML still outperforms VarDyn in reconstructing the top layer flow. Such results may indicates that a great component of the error conducted by the RG model can be explained by an advected field (whether only the sublayer geostrophic stream function or a more complex field). Rather than opposing SurfML to VarDyn, the (significant) remaining error may suggest that there would be some benefits in coupling the two method to overcome simplification-related errors.

6 Discussion

- As pointed out by Hurlburt, 1986, RG model can not account for bottom topography. SurfML however, could do so by introducing a topography-related term in the bottom layer equation (hence the regularization)
- Switch to real observation
- Investigate in details the benefits of mixing both VarDyn and SurfML methodology
- Improve model by choosing a more specific reduced-order space for $\tilde{\psi}_2$ to improve model reconstructions capacities

A Additional material regarding QG3L results

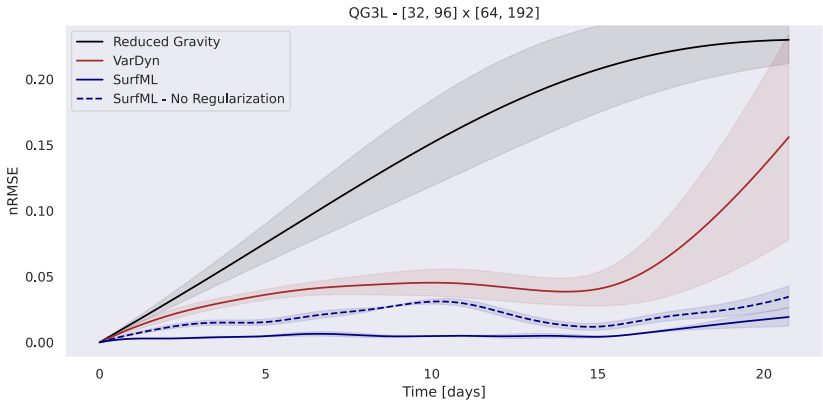

Figure 8: nRMSE on three-layer QG surface stream function (average over 3 consecutive 20-days periods) on Z1.

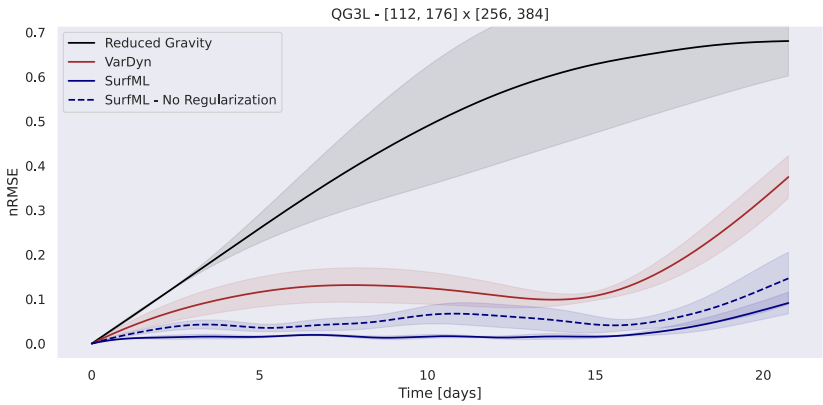

Figure 9: nRMSE on three-layer QG surface stream function (average over 3 consecutive 20-days periods) on Z4.

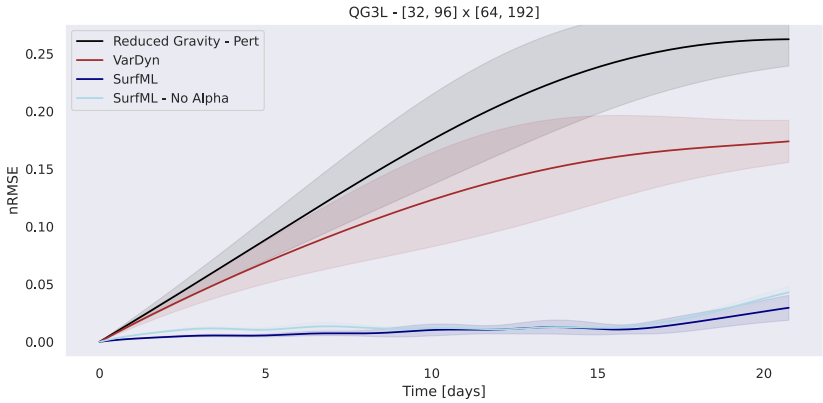

Figure 10: nRMSE on three-layer QG surface stream function (average over 3 consecutive 20-days periods) on Z1 using perturbed models.

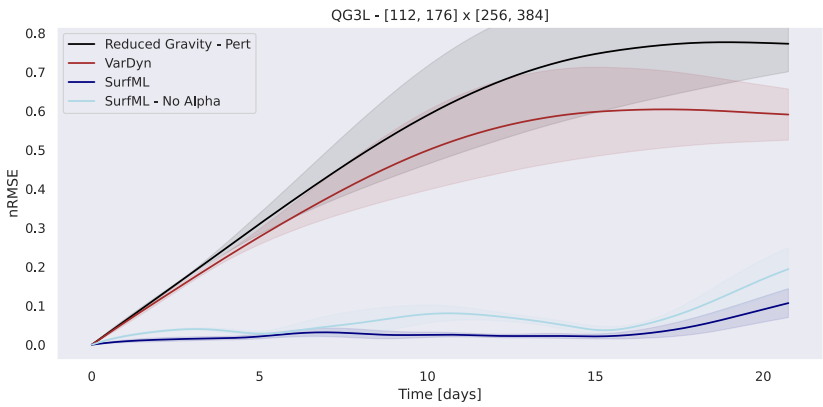

Figure 11: nRMSE on three-layer QG surface stream function (average over 3 consecutive 20-days periods) on Z4 using perturbed models.

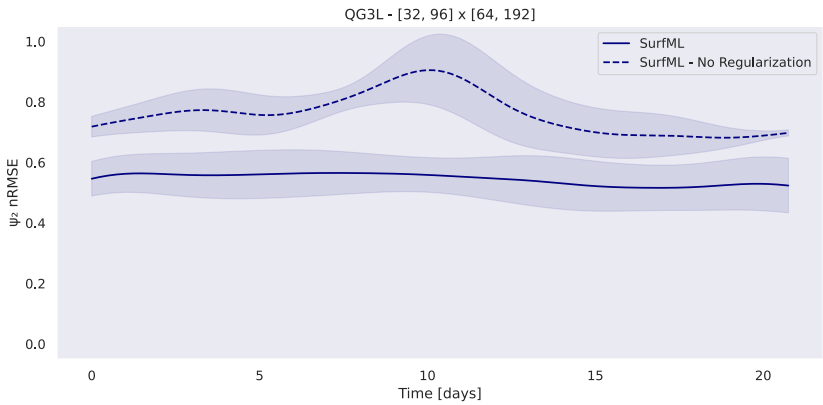

Figure 12: nRMSE on three-layer QG sublayer stream function (average over 3 consecutive 20-days periods) on Z1.

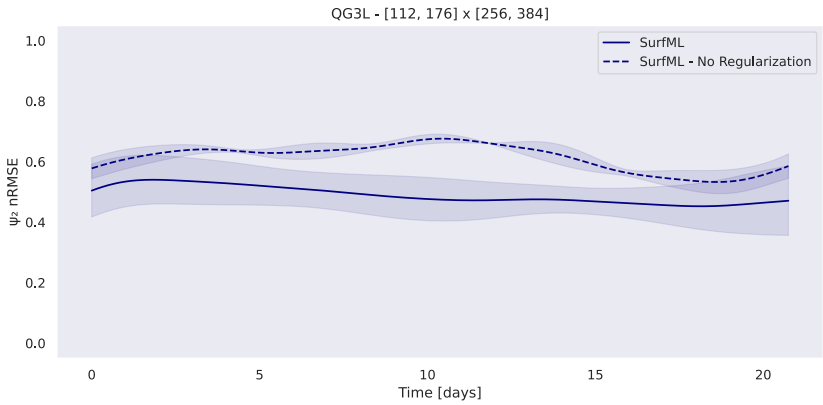

Figure 13: nRMSE on three-layer QG sublayer stream function (average over 3 consecutive 20-days periods) on Z4.

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